A Comprehensive Data-Driven Framework for Seller Lifetime Value Modeling and Behavioral Segmentation Using Probabilistic Models and Unsupervised Learning

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***Abstract*—E-commerce platforms increasingly depend on the sustained engagement of third-party sellers as a primary revenue source; however, existing Customer Lifetime Value (CLV) literature predominantly focuses on end consumers rather than the sellers themselves. This study addresses this gap by adapting and applying the BG/NBD and Gamma-Gamma probabilistic models, conventionally employed for buyer-side transaction modeling, to the seller population of a marketplace platform. Using the Brazilian E-Commerce Public Dataset by Olist, we construct a Seller Lifetime Value (SeLTV) framework that estimates individual seller activity probabilities and expected transaction volumes via the BG/NBD model, and approximates per-order monetary value through the Gamma-Gamma model, from which platform commission revenue is derived as a proxy for SeLTV. These probabilistic outputs are subsequently used as features for K-Means clustering, enabling the segmentation of sellers into behaviorally distinct cohorts with tailored retention strategies. An A/B testing protocol is further proposed to validate the causal impact of cohort-specific interventions. This work demonstrates the feasibility and interpretability of repurposing established buyer-side probabilistic models for seller-side lifetime value estimation, contributing a scalable analytical framework for seller relationship management on digital marketplace platforms.**

***Index Terms*—Seller Lifetime Value, BG/NBD, Gamma-Gamma, K-means, A/B Testing, Probabilistic Models, E-Commerce Analytics, Data-Driven Marketing**

I. Introduction

II. Exploratory Data Analysis

III. Methodology

A. Data Preprocessing and Feature Construction

1) Data Cleaning and Transaction Aggregation: This study utilizes the *Brazilian E-Commerce Public Dataset by Olist* to construct a seller-centric transactional database for Seller Lifetime Value (SeLTV) modeling. Since the objective of the study is to evaluate seller behavior from the perspective of a marketplace platform, the raw order-level and item-level datasets were integrated to create a unified seller-order transaction table.

The preprocessing stage began by merging the `orders`, `order_items`, and `sellers` tables using the shared identifiers `order_id` and `seller_id`. To ensure transactional consistency and economic validity, several filtering criteria were applied. Only transactions associated with delivered orders were retained, as these represent successfully completed commercial activities capable of generating realized platform revenue. Observations with missing seller identifiers, order identifiers, transaction timestamps, or product prices were removed. In addition, transactions with non-positive monetary values were excluded to avoid distortions in subsequent monetary estimation procedures.

Formally, a transaction record is retained only if it satisfies the following validity constraints:

$$\begin{aligned} \texttt{order\_status} &= \text{“delivered”} \\ \texttt{price} &> 0 \\ \texttt{seller\_id} &\neq \varnothing \\ \texttt{order\_id} &\neq \varnothing \\ \texttt{transaction\_date} &\neq \varnothing \end{aligned}$$

The transaction timestamp was defined using the order purchase datetime (`order_purchase_timestamp`), which serves as the temporal reference for all behavioral analyses. Gross Merchandise Value (GMV) was approximated using the product-level price field provided in the dataset.

To align the data structure with probabilistic customer-base modeling requirements, the cleaned item-level records were subsequently aggregated into seller-order transactions. Specifically, all items associated with the same seller and order were consolidated into a single transaction event. For each seller-order pair, the following aggregated variables were computed:

- Transaction date (minimum purchase timestamp)
- Total GMV
- Total freight value
- Number of items
- Seller geographic information (city and state)

The aggregation process transforms granular item-level observations into seller-level transactional sequences appropriate for Buy-Till-You-Die (BTYD) modeling frameworks such as BG/NBD and Gamma-Gamma. This transformation is particularly important because the BG/NBD model assumes discrete transaction occurrences over time at the entity level, which in this study corresponds to sellers operating on the marketplace platform.

2) Calibration and Holdout Split: To evaluate the predictive performance of the proposed probabilistic framework, the transaction data were partitioned into calibration and holdout periods following the conventional BTYD modeling paradigm.

The observation window begins on January 1, 2017 and includes all eligible transactions up to the maximum holdout horizon. The calibration period ends on April 30, 2018 at 23:59:59, while the holdout period extends beyond the calibration boundary for multiple forecasting horizons.

Let $t_c =$ April 30, 2018 denote the calibration cutoff date. Transactions satisfying:

$$\texttt{transaction\_date} \leq t_c$$

were assigned to the calibration dataset, whereas transactions occurring after the cutoff date were assigned to the holdout dataset:

$$t_c < \texttt{transaction\_date} \leq t_c + h$$

where $h \in \{30, 60, 90\}$ days represents the prediction horizon.

The calibration dataset is used for parameter estimation of the BG/NBD and Gamma-Gamma models, while the holdout dataset enables out-of-sample evaluation of transaction frequency and monetary value predictions. The use of multiple forecast horizons allows the robustness of the proposed framework to be examined across short-, medium-, and longer-term seller activity windows.

This temporal splitting strategy preserves the chronological structure of marketplace transactions and prevents information leakage from future observations into the training process.

3) Model Feature Construction: Following the calibration split, seller-level behavioral features were constructed to support probabilistic transaction modeling. The feature engineering procedure follows the standard Buy-Till-You-Die (BTYD) framework while adapting its interpretation to the seller-side marketplace context.

For each seller i, the following variables were computed from the calibration-period transactions:

a) Transaction Frequency: The BG/NBD model defines transaction frequency as the number of repeat transactions completed by a seller during the calibration period. Let:

$$x_i = n_i - 1 \tag{1}$$

where n_i denotes the total number of observed seller transactions. This definition excludes the first observed transaction because the BTYD framework models repeat purchasing behavior conditional on acquisition.

b) Recency: Recency measures the elapsed time between the seller's first and most recent observed transactions during the calibration period:

$$\text{Recency}_i = t_{i,\text{last}} - t_{i,\text{first}} \tag{2}$$

Recency captures the temporal concentration of seller activity and provides important information regarding seller engagement intensity.

c) Customer Age (T): The variable T_i represents the total observation duration between the seller's first transaction and the end of the calibration period:

$$T_i = t_c - t_{i,\text{last}} \tag{3}$$

where t_c denotes the calibration cutoff date. To ensure numerical stability during model estimation, extremely small values of T_i were constrained to a small positive constant.

d) Monetary Value: For Gamma-Gamma estimation, the monetary value variable was defined as the average GMV of repeat transactions only:

$$m_i = \frac{1}{x_i} \sum_{j=2}^{n_i} GMV_{ij} \tag{4}$$

This formulation is consistent with the assumptions of the Gamma-Gamma model, which estimates expected transaction value conditional on repeat purchasing behavior. Sellers without repeat transactions were assigned a monetary value of zero during preprocessing.

e) Additional Descriptive Features: Several supplementary variables were also retained for descriptive and downstream analytical purposes, including:

The following supplementary seller-level variables were also retained:

Total calibration-period orders
Total calibration-period GMV
Historical average order value (AOV)
First transaction date
Last transaction date

These seller-level features constitute the foundational inputs for subsequent BG/NBD activity forecasting and Gamma-Gamma monetary estimation procedures.

B. Seller Activity Modeling Using BG/NBD

The BG/NBD model, originally proposed by Fader, Hardie, and Lee [1], serves as the foundation for seller activity modeling in this framework. While the model was originally developed to predict repeat-purchase behavior of end consumers in non-contractual settings, we adapt and apply it to the seller population of the Olist marketplace, treating each seller as an independent entity whose transaction history with the platform is summarized by a Recency-Frequency-Tenure (R-F-T) triple.

1) BG/NBD Model Assumptions: The BG/NBD model is built upon five behavioral assumptions, restated here in the context of seller-side transactions:

a) Poisson transaction process: While a seller remains active, the number of fulfilled orders in a time period of length t follows a Poisson process with individual transaction rate λ. Equivalently, the inter-order time for an active seller is exponentially distributed with rate λ.

b) Gamma heterogeneity in transaction rates: Transaction rates λ vary across sellers according to a Gamma distribution with shape parameter r and scale parameter α:

$$f(\lambda \mid r, \alpha) = \frac{\alpha^r \lambda^{r-1} e^{-\lambda\alpha}}{\Gamma(r)}, \quad \lambda > 0 \tag{5}$$

c) Geometric dropout process: After each fulfilled order, a seller may permanently cease activity with probability p. The point of dropout is therefore geometrically distributed across transactions:

$$P(\text{inactive after } j\text{-th order}) = p(1-p)^{j-1}, \quad j = 1, 2, 3, \ldots \tag{6}$$

d) Beta heterogeneity in dropout probabilities: Dropout probabilities p vary across sellers according to a Beta distribution with shape parameters a and b:

$$f(p \mid a, b) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \quad 0 \le p \le 1 \tag{7}$$

e) Independence of λ and p: The transaction rate and dropout probability vary independently across sellers.

Prior to model fitting, a set of assumption checks is conducted on the calibration dataset to verify that the Olist seller transaction data is broadly consistent with the above conditions. These checks include validating the structural integrity of the R-F-T summary, examining the coefficient of variation of inter-order times to assess the Poisson process assumption, inspecting the frequency distribution for evidence of Gamma-consistent heterogeneity, and evaluating the stationarity of seller-level order rates across the early and late subperiods of calibration.

2) Expected Number of Orders Estimation: Each seller is represented by the summary tuple $(x,\, t_x,\, T)$, where x denotes the number of repeat orders observed in the calibration period $(0,\, T]$ and t_x is the time of the most recent order. The four model parameters $(r,\, \alpha,\, a,\, b)$ are estimated by maximizing the sample log-likelihood function:

$$\ell(r, \alpha, a, b) = \sum_{i=1}^{N} \ln L(r, \alpha, a, b \mid x_i, t_{x_i}, T_i) \tag{8}$$

where the individual-level likelihood is:

$$\begin{aligned} L(r, \alpha, a, b \mid x, t_x, T) &= \frac{B(a,\, b+x)}{B(a,b)} \frac{\Gamma(r+x)\,\alpha^r}{\Gamma(r)(\alpha+T)^{r+x}} \\ &+ \delta_{x>0} \frac{B(a+1,\, b+x-1)}{B(a,b)} \frac{\Gamma(r+x)\,\alpha^r}{\Gamma(r)(\alpha+t_x)^{r+x}} \end{aligned} \tag{9}$$

Optimization is performed using the `BetaGeoFitter` implementation from the `lifetimes` Python library. Once parameters are estimated, the expected number of future orders for seller i over a prediction horizon of length t is:

$$\begin{aligned} \mathbb{E}[Y(t) \mid x, t_x, T, r, \alpha, a, b] &= \frac{a+b+x-1}{a-1} \Bigg[1 - \left(\frac{\alpha+T}{\alpha+T+t} \right)^{r+x} \\ &\times {}_2F_1\left(r+x,\, b+x;\, a+b+x-1;\, \frac{t}{\alpha+T+t} \right) \Bigg] \\ &\Bigg/ \left[1 + \delta_{x>0} \frac{a}{b+x-1} \left(\frac{\alpha+T}{\alpha+t_x} \right)^{r+x} \right] \end{aligned} \tag{10}$$

where ${}_2F_1(\cdot)$ is the Gaussian hypergeometric function and $\delta_{x>0} = 1$ if $x > 0$, 0 otherwise. Predictions are generated across three forward-looking horizons $t \in \{30,\, 60,\, 90\}$ days to assess the sensitivity of downstream SeLTV estimates to the choice of prediction window.

In addition to forecasting future order volumes, the BG/NBD model yields the probability that a seller is still active (i.e., has not yet permanently dropped out) at the end of the calibration period T. This quantity is derived analytically from the model's posterior likelihood structure as:

$$\begin{aligned} &P(\text{active} \mid x, t_x, T, r, \alpha, a, b) \\ &= \frac{1}{1 + \delta_{x>0} \frac{a}{b+x-1} \left(\frac{\alpha+T}{\alpha+t_x} \right)^{r+x}} \end{aligned} \tag{11}$$

3) Seller Active Probability Derivation: In addition to the long-run activity status captured by Equation (11), a complementary forward-looking engagement probability is derived to quantify the likelihood that a seller places at least one order within a future horizon of length t. Given that the expected number of future orders $\mathbb{E}[Y_i(t)]$ is obtained from the BG/NBD model under a Poisson transaction process, the probability of observing at least one transaction in the period $(T, T+t]$ can be approximated as:

$$\hat{p}_i(t) = 1 - e^{-\mathbb{E}[Y_i(t)]} \tag{12}$$

This approximation follows directly from the zero-count probability of a Poisson random variable with mean $\mathbb{E}[Y_i(t)]$. Unlike $P(\text{active} \mid x, t_x, T)$, which reflects the posterior probability that a seller has not yet permanently churned at the end of the calibration period, $\hat{p}_i(t)$ is a horizon-conditional quantity that additionally accounts for the possibility that an active seller may simply not transact within the specified window.

4) Evaluation Metrics:

a) Holdout Target Construction: Model evaluation is conducted under a train-holdout split scheme. The calibration period spans from January 2017 through April 2018, during which the R-F-T summary for each seller is constructed. Three holdout windows of $t \in \{30, 60, 90\}$ days are then defined immediately following the calibration end. For each seller i and horizon t, the actual number of orders placed in the holdout window is computed as:

$$y_i^{(t)} = \left| \{ j : \text{order}_{ij} \text{ placed in } (T_{\text{cal}},\, T_{\text{cal}} + t] \} \right| \tag{13}$$

The binary activity indicator is then derived as:

$$a_i^{(t)} = \mathbf{1}\left[y_i^{(t)} > 0\right] \tag{14}$$

which equals 1 if seller i places at least one order during the holdout window, and 0 otherwise.

b) Order Volume Metrics: To assess the accuracy of the BG/NBD model in predicting the number of future orders $\hat{y}_i^{(t)} = \mathbb{E}[Y_i(t)]$ at the seller level, we compute the following metrics over the full evaluation population $\mathcal{S}$:

(i) Mean Absolute Error (MAE):

$$\text{MAE}_{\text{orders}}^{(t)} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left| \hat{y}_i^{(t)} - y_i^{(t)} \right| \tag{15}$$

(ii) Median Absolute Error (MedianAE):

$$\text{MedianAE}_{\text{orders}}^{(t)} = \operatorname*{median}_{i \in \mathcal{S}} \left| \hat{y}_i^{(t)} - y_i^{(t)} \right| \tag{16}$$

(iii) Weighted Absolute Percentage Error (WAPE):

$$\text{WAPE}_{\text{orders}}^{(t)} = \frac{\sum_{i \in \mathcal{S}} \left| \hat{y}_i^{(t)} - y_i^{(t)} \right|}{\sum_{i \in \mathcal{S}} y_i^{(t)}} \tag{17}$$

WAPE is preferred over MAPE in this setting because many sellers have $y_i^{(t)} = 0$, which renders percentage-based errors undefined at the individual level. WAPE instead measures the total absolute prediction error as a fraction of the total actual order volume across the seller population, yielding a pooled, volume-weighted error rate.

c) Seller Activity Classification Metrics: The expected number of future orders $\hat{y}_i^{(t)}$ is used as a continuous ranking score to evaluate the model's ability to identify which sellers will remain active during the holdout period. Rather than applying a fixed classification threshold, we treat the problem as a ranking task and evaluate using:

(i) ROC-AUC: The area under the Receiver Operating Characteristic curve measures the probability that a randomly selected active seller is ranked higher than a randomly selected inactive seller by the model's scoring function.

$$\text{ROC-AUC}^{(t)} = P\left(\hat{y}_i^{(t)} > \hat{y}_j^{(t)} \;\middle|\; a_i^{(t)} = 1,\ a_j^{(t)} = 0\right) \tag{18}$$

(ii) PR-AUC: The area under the Precision-Recall curve measures ranking quality under class imbalance, placing greater emphasis on the model's ability to correctly rank truly active sellers at the top of the list. PR-AUC is particularly relevant in contexts where the active seller population is smaller than the inactive population.

(iii) Lift at Top-k: For $k \in \{0.10, 0.20\}$, the lift at top-k measures the enrichment of active sellers among the top k fraction of sellers ranked by $\hat{y}_i^{(t)}$ relative to the population base rate:

$$\text{Lift}_k^{(t)} = \frac{\frac{1}{|\mathcal{S}_k|} \sum_{i \in \mathcal{S}_k} a_i^{(t)}}{\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} a_i^{(t)}} \tag{19}$$

where $\mathcal{S}_k$ denotes the top-k fraction of sellers ordered by descending predicted order count. A lift of L at top-10% indicates that focusing on the top-ranked 10% of sellers yields L times the active seller density compared to random selection.

C. Monetary Value Modeling Using Gamma-Gamma

1) Gamma-Gamma Model Assumptions: To estimate the expected monetary value of each seller's future transactions, we employ the Gamma-Gamma model of spend per transaction as presented by Fader and Hardie [3]. The model characterizes the cross-sectional distribution of average transaction values under three core assumptions:

a) Gamma-distributed transaction values: The monetary value z_i of each individual order placed by a seller is assumed to follow a Gamma distribution with shape parameter p and rate parameter ν, yielding a latent mean transaction value $\zeta = p/\nu$. For a seller with x repeat orders, the observed average order value $\bar{z} = \sum_{i=1}^{x} z_i / x$ follows a $\text{Gamma}(px,\, \nu x)$ distribution.

b) Gamma heterogeneity in mean transaction values: The latent mean transaction value ζ varies across sellers through a Gamma mixing distribution on ν, i.e., $\nu \sim \text{Gamma}(q, \gamma)$, giving rise to an inverse-Gamma marginal distribution of ζ with mean:

$$\mathbb{E}[Z \mid p, q, \gamma] = \frac{p\gamma}{q - 1} \tag{20}$$

c) Independence of monetary value and transaction process: The distribution of average transaction values across sellers is assumed to be independent of the underlying transaction process. This assumption is empirically verified prior to model fitting by examining the Pearson and Spearman correlations between seller repeat frequency and average order value, both on raw and log-transformed scales.

Prior to model fitting, assumption checks are conducted to verify that the monetary value data satisfies the structural requirements of the model. These checks include confirming the absence of non-positive monetary values, assessing the right-skewed shape of the AOV distribution consistent with a Gamma form, verifying the independence between frequency and monetary value, and evaluating the temporal stability of seller-level AOV across the calibration period.

2) Expected Average Order Value Estimation: The Gamma-Gamma model parameters $(p,\, q,\, \gamma)$ are estimated by maximizing the sample log-likelihood:

$$\ell(p, q, \gamma) = \sum_{i=1}^{I} \ln f(\bar{z}_i \mid p, q, \gamma;\, x_i) \tag{21}$$

where the marginal density of $\bar{z}$ given x repeat transactions is:

$$f(\bar{z} \mid p, q, \gamma;\, x) = \frac{\Gamma(px + q)}{\Gamma(px)\,\Gamma(q)} \cdot \frac{\bar{z}^{\,px-1}\, x^{px}\, \gamma^{q}}{(\gamma + x\bar{z})^{px+q}} \tag{22}$$

Estimation is restricted to sellers with at least one repeat transaction ($x > 0$) and a positive monetary value, using the `GammaGammaFitter` implementation from the `lifetimes` library. For sellers ineligible for Gamma-Gamma estimation (i.e., one-time sellers with $x = 0$), the conditional expectation reduces to the population mean $\mathbb{E}[Z]$, and their observed historical AOV is used as a heuristic fallback.

Once the parameters are estimated, the posterior expected average order value for each seller is computed via Bayesian shrinkage toward the population mean:

$$\begin{aligned}\mathbb{E}[Z \mid \bar{z}, x, p, q, \gamma] \\ = \frac{p(\gamma + x\bar{z})}{px + q - 1} \\ = \left(\frac{q-1}{px+q-1}\right)\frac{p\gamma}{q-1} + \left(\frac{px}{px+q-1}\right)\bar{z}\end{aligned} \tag{23}$$

This expression is a convex combination of the population mean and the seller's own observed AOV, where the weight placed on the individual observation increases monotonically with the number of transactions x. This shrinkage property is particularly well suited for sparse transaction histories, which are prevalent in the Olist seller population.

3) Seller Lifetime Value Derivation: Seller Lifetime Value (SeLTV) is defined as the total platform commission revenue attributable to a given seller over a future prediction horizon t. Combining the outputs of the BG/NBD and Gamma-Gamma models under the independence assumption between transaction timing and monetary value [1], SeLTV is formulated as:

$$\mathrm{SeLTV}_i(t) = \mathbb{E}[Y_i(t)] \times \mathbb{E}[Z_i \mid \bar{z}_i, x_i] \times r_c \tag{24}$$

where $\mathbb{E}[Y_i(t)]$ is the expected number of future orders from Equation (10), $\mathbb{E}[Z_i \mid \bar{z}_i, x_i]$ is the Bayesian-adjusted expected AOV from Equation (23), and r_c is the platform's commission rate which is the share of GMV retained as revenue by the marketplace.

This formulation differs fundamentally from conventional buyer-side CLV in that the unit of value is not consumer expenditure but platform commission revenue generated by seller activity. The commission-based proxy is justified on the grounds that marketplace revenue is structurally proportional to seller-generated GMV, obviating the need for a per-category margin model. Furthermore, by grounding SeLTV in probabilistic estimates of both future order volume and expected monetary value, the framework naturally propagates uncertainty from both model components into the final value estimate, enabling more nuanced downstream segmentation than a purely historical heuristic would permit.

4) Evaluation Metrics:

a) Holdout Target Construction: For each seller i and holdout horizon t, the actual average order value (AOV) is computed exclusively over sellers who placed at least one order during the holdout window, i.e., sellers with $y_i^{(t)} > 0$:

$$\bar{z}_i^{(t)} = \frac{\sum_{j=1}^{y_i^{(t)}} z_{ij}^{(t)}}{y_i^{(t)}} = \frac{G_i^{(t)}}{y_i^{(t)}} \tag{25}$$

where $G_i^{(t)}$ is the total GMV generated by seller i during the holdout window and $z_{ij}^{(t)}$ is the value of the j-th individual order. The restriction to sellers with $y_i^{(t)} > 0$ is necessary because AOV is undefined for sellers who generate no orders in the holdout period.

Similarly, the actual commission generated by seller i over horizon t is defined as:

$$c_i^{(t)} = G_i^{(t)} \times r_c \tag{26}$$

and the predicted commission is:

$$\hat{c}_i^{(t)} = \hat{y}_i^{(t)} \times \mathbb{E}[Z_i \mid \bar{z}_i, x_i] \times r_c \tag{27}$$

b) AOV Accuracy Metrics: The Gamma-Gamma model is evaluated on its ability to estimate the expected AOV $\mathbb{E}[Z_i \mid \bar{z}_i, x_i]$ for each seller. Evaluation is conducted on the subset of sellers with holdout orders $y_i^{(t)} > 0$, denoted $\mathcal{S}_+^{(t)} = \{i : y_i^{(t)} > 0\}$:

(i) MAE (AOV):

$$\mathrm{MAE}_{\mathrm{AOV}}^{(t)} = \frac{1}{|\mathcal{S}_+^{(t)}|} \sum_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right| \tag{28}$$

(ii) Median AE (AOV):

$$\mathrm{MedianAE}_{\mathrm{AOV}}^{(t)} = \operatorname*{median}_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right| \tag{29}$$

(iii) WAPE (AOV):

$$\mathrm{WAPE}_{\mathrm{AOV}}^{(t)} = \frac{\sum_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right|}{\sum_{i \in \mathcal{S}_+^{(t)}} \bar{z}_i^{(t)}} \tag{30}$$

(iv) Spearman Rank Correlation (AOV): Since the primary use of AOV estimates is to rank sellers by expected monetary value, we additionally compute the Spearman rank correlation between predicted and actual AOV:

$$\rho_{\mathrm{AOV}}^{(t)} = \mathrm{Spearman}\Big(\mathbb{E}[Z_i],\ \bar{z}_i^{(t)}\Big), \quad i \in \mathcal{S}_+^{(t)} \tag{31}$$

A high Spearman correlation indicates that the model correctly orders sellers by their monetary value, even if absolute estimates are imperfect which is a property that is sufficient for the downstream segmentation task.

c) Commission Accuracy Metrics: The final SeLTV estimate $\hat{c}_i^{(t)}$ integrates both model components. Its accuracy is evaluated across the full seller population $\mathcal{S}$, using the same family of metrics applied to orders:

(i) MAE and Median AE (commission): Point-error measures at the individual seller level, analogous to the order-level definitions above.

(ii) WAPE (commission):

$$\mathrm{WAPE}_{\mathrm{comm}}^{(t)} = \frac{\sum_{i\in\mathcal{S}} \left|\hat{c}_i^{(t)} - c_i^{(t)}\right|}{\sum_{i\in\mathcal{S}} c_i^{(t)}} \tag{32}$$

(iii) Spearman Rank Correlation (commission): The Spearman rank correlation between predicted and actual commission evaluates whether the model correctly prioritizes sellers by their revenue contribution to the platform:

$$\rho_{\mathrm{comm}}^{(t)} = \mathrm{Spearman}\Big(\hat{c}_i^{(t)},\ c_i^{(t)}\Big), \quad i \in \mathcal{S} \tag{33}$$

This metric is central to evaluating the usefulness of SeLTV for marketing resource allocation, since a high rank correlation implies that directing effort toward high-SeLTV sellers reliably captures a disproportionate share of actual platform commission.

D. Seller Segmentation Using K-Means Clustering

1) Feature Selection, Scaling, and Transformation:

a) Feature Selection: The feature set for clustering is constructed exclusively from the probabilistic outputs of the BG/NBD and Gamma-Gamma models. The prediction horizon used as input to the segmentation stage will be selected based on the model evaluation results: specifically, the horizon that yields the best trade-off across order volume accuracy, seller activity classification quality, and commission ranking performance is adopted as the basis for all downstream clustering features. Three features are constructed at the chosen horizon t^*:

(i) $\hat{y}_i^{(t^*)}$: the expected number of orders over the selected horizon, derived from the BG/NBD conditional expectation (Equation (10)).

(ii) $\hat{p}_i^{(t^*)}$: an approximate probability that seller i places at least one order within the horizon, defined as:

$$\hat{p}_i^{(t^*)} = 1 - e^{-\hat{y}_i^{(t^*)}} \tag{34}$$

This Poisson-based approximation serves as a continuous activity score, capturing the seller's likelihood of remaining engaged with the platform.

(iii) $\hat{c}_i^{(t^*)}$: the expected commission revenue attributable to seller i over the selected horizon, as defined in Equation (24).

These three features jointly encode a seller's expected activity intensity ($\hat{y}_i^{(t^*)}$), engagement probability ($\hat{p}_i^{(t^*)}$), and revenue potential ($\hat{c}_i^{(t^*)}$) which are the three dimensions most relevant to platform-level seller management decisions. Importantly, all three are forward-looking model outputs rather than raw historical statistics, ensuring that the resulting segments reflect predicted future behavior rather than past behavior alone.

b) Transformation and Standardization: The raw feature distributions are highly right-skewed, as a small number of high-frequency sellers with large transaction volumes dominate all three features. Applying K-Means directly to such distributions would cause centroids to be heavily pulled toward outliers, yielding clusters that reflect scale effects rather than genuine behavioral differences.

To address this, we apply the Yeo-Johnson power transformation [6] followed by standardization to zero mean and unit variance, using the `PowerTransformer` from `scikit-learn` with `standardize=True`. The Yeo-Johnson family is chosen over the Box-Cox transformation because it accommodates zero-valued observations (i.e., sellers with $\hat{y}_i^{(t^*)} \approx 0$), which are prevalent in the dataset. Formally, for a feature value v, the transformation is:

$$\psi_\lambda(v) = \begin{cases} \dfrac{(v+1)^\lambda - 1}{\lambda}, & \lambda \neq 0,\ v \geq 0 \\ \ln(v+1), & \lambda = 0,\ v \geq 0 \\ -\dfrac{(-v+1)^{2-\lambda} - 1}{2-\lambda}, & \lambda \neq 2,\ v < 0 \\ -\ln(-v+1), & \lambda = 2,\ v < 0 \end{cases} \tag{35}$$

where the optimal λ is estimated via maximum likelihood separately for each feature. The transformed features are then standardized to produce the final input matrix $\mathbf{X}_{\text{scaled}} \in \mathbb{R}^{N\times 3}$, on which K-Means is fitted.

2) Determination of Optimal Number of Clusters: Selecting the appropriate number of clusters K requires balancing intra-cluster compactness against inter-cluster separation. We employ a two-stage procedure: (i) the Elbow method to identify a candidate range of K, followed by (ii) the Silhouette criterion to select the optimal value within that range.

a) Stage 1 - Elbow Method: K-Means is fitted for $K \in \{2, 3, \ldots, K_{\max}\}$ and the within-cluster sum of squared distances (inertia) is recorded for each:

$$\mathcal{J}(K) = \sum_{k=1}^{K} \sum_{i\in\mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \tag{36}$$

where $\mathcal{C}_k$ is the set of sellers assigned to cluster k and $\boldsymbol{\mu}_k$ is the corresponding centroid. As K increases, $\mathcal{J}(K)$ decreases monotonically; however, the rate of decrease typically exhibits a pronounced "elbow" that is a point of inflection beyond which additional clusters yield diminishing marginal reduction in inertia. The elbow chart is used to identify a range of plausible values of K that meaningfully reduce $\mathcal{J}$, rather than selecting a single K mechanically.

b) Stage 2 - Silhouette Analysis: For each candidate K identified in Stage 1, we compute the mean Silhouette score across all sellers. For seller i, the Silhouette coefficient is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\ b(i)\}} \tag{37}$$

where $a(i)$ is the mean intra-cluster distance from seller i to all other members of its assigned cluster, and $b(i)$ is the mean distance from seller i to all sellers in the nearest neighboring cluster:

$$a(i) = \frac{1}{|\mathcal{C}_{k(i)}| - 1} \sum_{j \in \mathcal{C}_{k(i)},\, j \neq i} \|\mathbf{x}_i - \mathbf{x}_j\|$$
$$b(i) = \min_{k \neq k(i)} \frac{1}{|\mathcal{C}_k|} \sum_{j \in \mathcal{C}_k} \|\mathbf{x}_i - \mathbf{x}_j\| \tag{38}$$

The coefficient $s(i) \in [-1, 1]$, where values close to $+1$ indicate that seller i is well-matched to its own cluster and clearly separated from neighboring clusters, values near 0 indicate boundary membership, and negative values suggest possible misassignment. The mean Silhouette score over the full population:

$$\bar{s}(K) = \frac{1}{N} \sum_{i=1}^{N} s(i) \tag{39}$$

is used as the primary selection criterion: the value of K that maximizes $\bar{s}(K)$ within the candidate range identified by the elbow chart is chosen as the final number of clusters.

c) Final Configuration: With the optimal number of clusters K being selected based on the two-stage procedure described above, the final K-Means model is fitted on $\mathbf{X}_{\text{scaled}}$ with `n_init` $= 20$ random initializations and `max_iter` $= 300$ iterations per run, using `random_state` $= 42$ for reproducibility. The solution with the lowest inertia across all initializations is retained as the final clustering.

3) Cluster Interpretation: The K-Means clustering framework successfully identified three distinct seller segments characterized by substantial differences in transactional behavior, activity probability, and projected commercial contribution. By integrating probabilistic outputs derived from the BG/NBD and Gamma-Gamma models, specifically expected future orders, estimated seller survival probability, and projected commission value over a 60-day horizon, the proposed framework provides a behaviorally grounded approach to seller segmentation beyond traditional descriptive clustering techniques.

The resulting segmentation reveals a highly concentrated commercial structure within the marketplace ecosystem and divides the cluster into 3 types.

a) High-Value Active Sellers: Cluster 2 represents the platform's most commercially valuable seller segment. Sellers in this cluster exhibit extremely high activity probability, with median purchase probability approaching 99.9%, alongside the highest projected transaction frequency across the seller population. According to the BG/NBD framework, these sellers can be considered behaviorally "alive," showing minimal signs of churn or declining engagement.

The cluster also generates substantially higher projected commission values than all other segments. However, the large gap between mean and median values for expected orders and commission indicates a strongly right-skewed distribution, suggesting the presence of top-tier sellers contributing disproportionately high transaction volumes and revenue. Such long-tail dynamics are common in marketplace ecosystems where a small subset of elite sellers drives most platform activity.

From a business perspective, Cluster 2 functions as the platform's primary revenue engine. Consequently, even moderate increases in churn within this segment could significantly impact short-term commission performance. The strong internal heterogeneity further suggests that additional sub-segmentation may be beneficial to distinguish ultra-high-value sellers from moderately active sellers.

Strategically, this cluster should receive the highest operational priority through retention-focused initiatives such as dedicated seller relationship management, loyalty programs, premium visibility support, and early-warning systems for churn detection.

b) Mid-Tier Engaged Sellerss: Cluster 1 represents a transitional seller segment characterized by moderate activity probability but relatively low transaction volume. Although expected order frequency remains limited, sellers within this group still maintain reasonably high survival probability, with median purchase probability exceeding 70%. This indicates that the segment consists not of fully churned sellers, but of sellers positioned between stable engagement and gradual disengagement.

Compared to Cluster 2, Cluster 1 exhibits a more homogeneous behavioral structure, with limited evidence of extreme outliers. Sellers continue interacting with the platform, yet their transaction frequency remains relatively low. This suggests that the primary issue may not be inactivity itself, but operational friction affecting conversion performance.

Possible explanations include onboarding difficulties, insufficient marketplace visibility, weak product discoverability, seasonal demand fluctuations, or competition from alternative platforms.

From a strategic perspective, Cluster 1 represents the platform's most promising growth opportunity. Activating and scaling existing sellers is generally more cost-effective than acquiring new ones, making this segment highly valuable for targeted intervention strategies. Recommended actions include onboarding optimization, activation campaigns, personalized incentives, and seller coaching initiatives.

c) Dormant Sellers: Cluster 0 consists of sellers with extremely weak future activity signals and minimal projected commercial contribution. The BG/NBD model assigns very low survival probabilities to this segment, with median purchase probability below 10%, indicating that most sellers are behaviorally close to churn.

Expected transaction activity within this cluster is nearly negligible, and unlike Cluster 2, the segment demonstrates relatively limited internal heterogeneity. Despite accounting for more than one-third of all sellers, Cluster 0 contributes only a marginal share of projected commission.

Therefore, the primary concern associated with this segment is not direct revenue loss, but operational inefficiency and opportunity cost. Resources allocated toward these sellers may generate greater returns if redirected toward higher-potential segments such as Clusters 1 and 2.

Nevertheless, not all sellers in this cluster should be considered permanently inactive. Some may exhibit seasonal selling behavior, meaning temporary inactivity does not necessarily imply permanent churn. Additional temporal analysis is therefore necessary before aggressive offboarding decisions are made.

Operationally, this segment is best managed through lightweight and scalable interventions such as automated reactivation campaigns, low-cost reminders, and inactive monitoring systems.

IV. Experimental Design

V. Results

A. Data Assumption Validation

Prior to model fitting, a structured series of assumption checks is conducted on the calibration dataset comprising 2,242 sellers, spanning January 2017 through April 2018. The checks are organized into two groups corresponding to the BG/NBD and Gamma-Gamma models respectively.

1) BG/NBD Assumption Checks:

a) Poisson Transaction Process: The Poisson process assumption requires that inter-order times follow an exponential distribution, which implies a coefficient of variation (CV) of inter-order gaps equal to approximately 1.0 at the individual seller level.

The analysis was restricted to sellers with at least five repeat transactions ($N = 1{,}213$ sellers) to ensure reliable estimation of inter-order gap variability. The median CV of inter-order gaps is 1.16 and the mean CV is 1.25, both in close proximity to the theoretical value of 1.0. The P10-P90 range spans 0.79 to 1.81, indicating that the assumption is broadly satisfied across the seller population. While 49.2% of inter-order gaps are shorter than one day, this reflects genuine multi-order behavior within a marketplace setting rather than a data artifact, as transactions have already been consolidated at the seller-order level.

b) Gamma Heterogeneity in Transaction Rates and Beta Heterogeneity in Dropout Probabilities: The Gamma mixing distribution requires that transaction rates exhibit substantial heterogeneity across sellers, typically manifested as a strongly right-skewed frequency distribution. The calibration sample of 2,242 sellers displays exactly this pattern: the mean frequency of 30.9 orders far exceeds the median of 6, with the distribution extending to a maximum of 1,515 (Table I). The concentration analysis provides further corroborating evidence: the top 1% of sellers (22 sellers) account for 24.3% of total orders, while the top 10% account for 66.5% (Table II), consistent with a Pareto-like dispersion arising from Gamma-distributed transaction rates rather than domination by a small group of extreme outliers.

In addition, according to Table I, the co-existence of a large group of low-frequency sellers (median $= 6$, P25 $= 1$) alongside a long-tailed group of high-frequency sellers (P99 $= 398$, max $= 1{,}515$) also implies that dropout probabilities are widely dispersed across the population, consistent with a Beta distribution over $(0, 1)$ rather than a degenerate point mass. Sellers with high estimated dropout probability contribute predominantly to the lower tail of the frequency distribution, while those with low dropout probability populate the upper tail.

TABLE I
Seller Repeat Order Frequency Distribution (Calibration Period)

Statistic	Value
Total sellers	2,242
Repeat sellers ($x > 0$)	1,828 (81.5%)
One-time sellers	414 (18.5%)
Mean frequency	30.9
Median frequency	6
P90 frequency	70
P99 frequency	398
Max frequency	1,515

TABLE II
Seller Transaction Concentration (Calibration Period)

Top %	N Sellers	Order Share	GMV Share
1%	22	24.3%	24.6%
5%	112	51.4%	51.5%
10%	224	66.5%	65.7%

c) Geometric Dropout Process: The geometric dropout assumption states that sellers can permanently cease activity immediately after any transaction, with a constant per-transaction dropout probability p. Although the assumption cannot be tested directly from aggregate statistics, the observed proportion of one-time sellers provides empirical evidence consistent with the underlying mechanism. Specifically, 18.5% of sellers completed only a single transaction during the calibration period, while 81.5% generated at least one repeat order. The presence of a substantial one-time seller group is consistent with the geometric dropout mechanism, under which some sellers disengage immediately after their first transaction with the platform.

d) Independence of λ and p: The independence between transaction rate and dropout probability is a structural assumption of the BG/NBD model that cannot be directly tested from observed data alone. Its validity is evaluated indirectly through the stationarity check. As reported in the stationarity check below, the median late-to-early order rate ratio of 1.39 is driven primarily by marketplace-level growth rather than a selective pattern in which high-rate sellers also exhibit systematically different dropout behavior, providing no evidence against the independence assumption.

Stationarity of Transaction Rates The BG/NBD model further requires that each seller's transaction rate λ remain approximately constant over time. Marketplace-level diagnostics reveal a clear growth trend: orders per active seller increased from approximately 3.4 in January 2017 to 7.5 in January 2018, a approximately two-time increase over the calibration window. At the individual seller level, the median late-to-early order rate ratio is 1.39, indicating a mild but non-negligible upward drift. This constitutes a soft violation of the stationarity assumption attributable to platform-wide buyer growth rather than intrinsic seller behavior change. The resulting bias is expected to manifest as a slight underestimation of future order volumes, and is considered manageable given the 16-month calibration window over which the model learns a time-averaged transaction rate.

2) Gamma-Gamma Model Assumptions:

a) Gamma-Distributed Transaction Values: The Gamma distributional assumption requires that monetary values be strictly positive and that the cross-sectional distribution of seller-level AOV be right-skewed, with most sellers concentrated at lower values and a long upper tail. Both conditions are satisfied in the calibration data. No zero or negative monetary values are present among the eligible sellers. The AOV distribution exhibits a skewness of 10.58 and a median of BRL 106.38 substantially below the mean of BRL 190.60, consistent with the expected Gamma shape (Table III). Figure 1 further illustrates the strongly right-skewed distribution of seller-level AOV values.

TABLE III
SELLER AVERAGE ORDER VALUE DISTRIBUTION (CALIBRATION PERIOD, $N = 1{,}828$)

Statistic	Value (BRL)
Min	3.70
P25	61.05
Median	106.38
Mean	190.60
P75	189.17
P95	640.52
Max	8,370.00
Std	340.70
Skewness	10.58
Excess kurtosis	201.73
Non-positive values	0

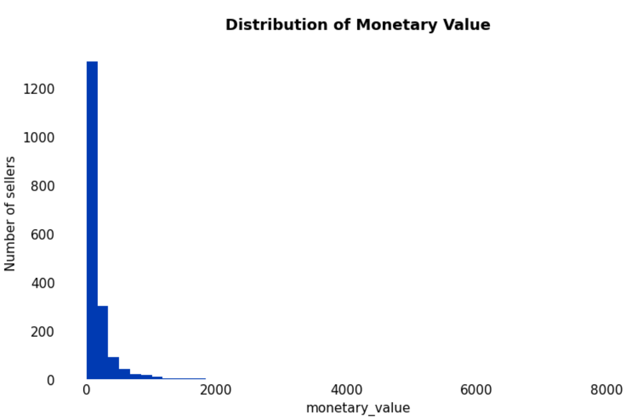

Fig. 1. Distribution of seller-level AOV in the calibration period.

b) Gamma Heterogeneity in Mean Transaction Values: The Gamma mixing distribution on latent mean AOV requires that seller-level spending levels vary substantially across the population which is a condition clearly satisfied by the distributional statistics above. The ratio of standard deviation (340.70) to mean (190.60) exceeds 1.78, and the P95-to-median ratio of approximately 6.0 confirms the degree of cross-sectional dispersion in seller spending that the inverse-Gamma marginal distribution is designed to capture. The 81.5% model eligibility rate further ensures that parameter estimation is conducted on a large and representative sample rather than a select high-frequency subgroup.

c) Independence of Monetary Value and Transaction Process: The independence assumption is assessed through four correlation measures between seller repeat frequency and average order value, on both raw and log-transformed scales (Table IV). All four metrics fall well below established thresholds, with the Spearman correlation being effectively zero at $+0.005$ on both scales. The extended Spearman correlation matrix confirms that monetary value is orthogonal to all R-F-T dimensions: its correlations with recency ($+0.054$) and tenure (-0.003) are negligible. These results provide exceptionally strong empirical support for the independence assumption.

TABLE IV
FREQUENCY-MONETARY VALUE INDEPENDENCE CHECK

Scale	Metric	Value	Threshold
Raw	Pearson	-0.062	$\lvert r\rvert < 0.15$
Raw	Spearman	$+0.005$	$\lvert r\rvert < 0.15$
Log-log	Pearson	-0.033	$\lvert r\rvert < 0.20$
Log-log	Spearman	$+0.005$	$\lvert r\rvert < 0.20$

Stationarity of Monetary Value The temporal stability of seller AOV is evaluated on the 655 sellers with at least two transactions in both the early (before September 2017) and late (after September 2017) subperiods. The Pearson correlation between early-period and late-period AOV is 0.76, confirming that seller-level AOV rankings are well preserved over time (Figure 2).

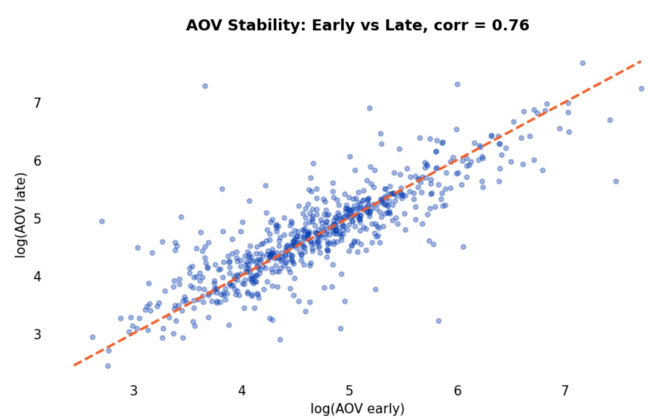

Fig. 2. Scatter plot of log(AOV) early vs. late periods

B. Model Performance Assessment

1) Evaluation of the BG/NBD Transaction and Activity Model:

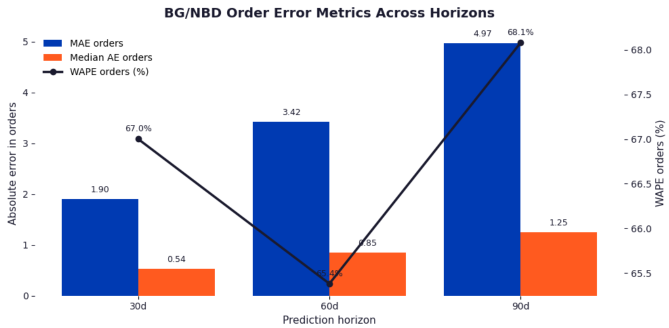

Fig. 3. Error metrics accross 3 horizons of BG/NBD model

a) Order Volume Accuracy: Figure 3 shows that absolute error increases monotonically with horizon length, with MAE rising from 1.90 at 30 days to 3.42 at 60 days and 4.97 at 90 days. This is an expected pattern, as cumulative order counts grow with the length of the prediction window. More informatively, WAPE remains relatively stable across horizons, ranging from 65.4% to 67.8%, indicating that the proportional error at the seller level does not deteriorate substantially as the horizon extends. The persistently elevated WAPE reflects the intrinsic difficulty of point-forecasting individual seller order volumes in a setting characterized by high behavioral heterogeneity.

b) Seller Activity Classification: The model demonstrates stronger performance in distinguishing active from inactive sellers than in absolute order count forecasting. PR-AUC improves from 0.865 at 30 days to 0.906 at 60 days and 0.915 at 90 days, while ROC-AUC remains stable at approximately 0.90 across all horizons (Figure 4), indicating robust and consistent ranking of sellers by their activity probability.

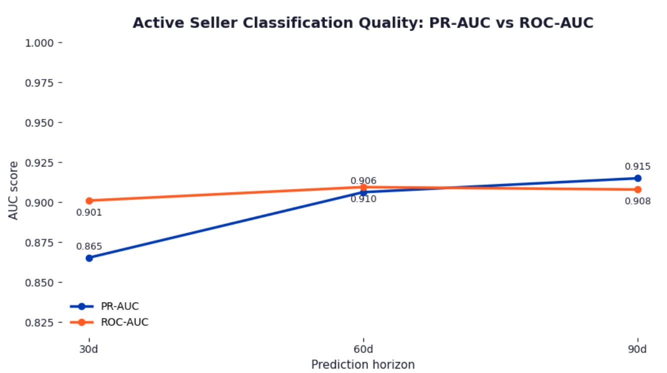

Fig. 4. Active Seller Classification Quality: PR-AUC vs. ROC-AUC

The lift analysis further confirms this: Lift@10 of 2.34× at 30 days indicates that concentrating attention on the top-decile sellers ranked by predicted order volume yields an active seller density 2.34 times higher than random selection. Lift@10 declines to 2.00× and 1.87× at 60 and 90 days respectively (Figure 5), a natural consequence of a larger proportion of sellers becoming active over longer horizons, which reduces the discriminative advantage of targeting the top decile.

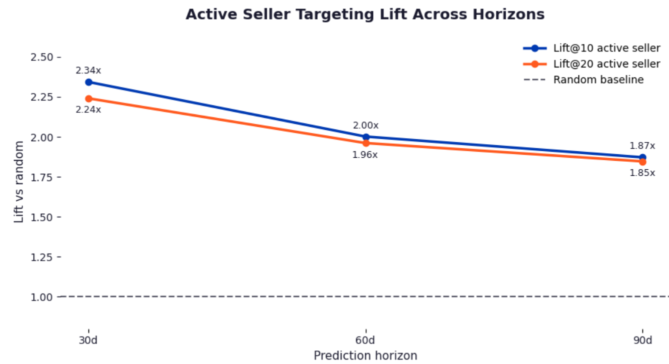

Fig. 5. Active Seller Targeting Lift Across Horizons

These results suggest that the BG/NBD model is best suited for seller activity ranking and prioritization rather than precise individual-level order count forecasting.

2) Evaluation of the Gamma-Gamma Monetary Model:

a) AOV Point Estimation Accuracy: In contrast to BG/NBD, Gamma-Gamma metrics improve as the horizon extends. WAPE on AOV decreases from 43.3% at 30 days to 41.6% at 60 days and 38.3% at 90 days, and MAE declines from 75.98 to 72.05 over the same range (Figure 6). This improvement occurs because longer holdout windows contain more transactions per seller, causing the observed AOV to converge toward the seller's true latent mean which is the quantity the Gamma-Gamma model is designed to estimate. The persistent gap between MAE and Median AE across all horizons reflects the right-skewed AOV distribution: a small segment of sellers with unusually high-value orders inflates the mean absolute error, and precise point estimation for this segment should not be expected.

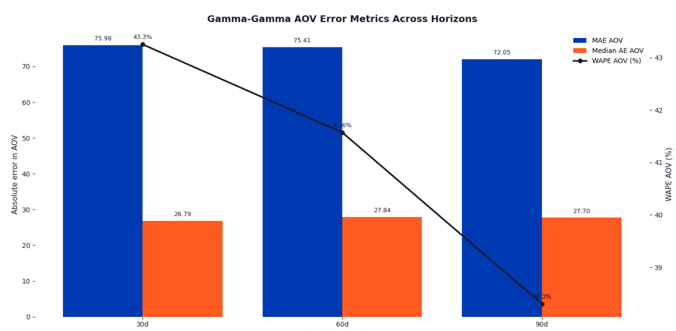

Fig. 6. Active Seller Targeting Lift Across Horizons

b) AOV Ranking Quality: The more operationally relevant capability of the Gamma-Gamma model lies in its ability to rank sellers by expected monetary value. Spearman rank correlation between predicted and actual AOV is stable and high across all horizons, ranging from 0.744 at 30 days to 0.769 at 90 days (Figure 7). This indicates that the model reliably preserves the relative ordering of sellers by spending level, even where absolute estimates are imprecise. For a marketplace operator, this ranking capability is of direct strategic value: it enables the platform to identify which sellers generate disproportionate per-order revenue independently of their transaction frequency.

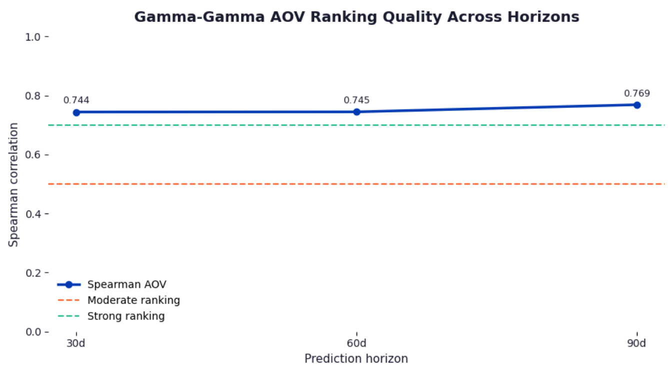

Fig. 7. Active Seller Targeting Lift Across Horizons

3) Integrated SeLTV Estimation Performance: Table V reports the evaluation metrics for the combined BG/NBD and Gamma-Gamma pipeline on the expected commission target.

TABLE V
COMBINED MODEL (EXPECTED COMMISSION) EVALUATION METRICS

Metric	**30 days**	**60 days**	**90 days**
MAE (commission)	44.93	77.36	111.32
Median AE (commission)	10.22	17.66	26.64
WAPE (commission)	0.753	0.708	0.737
Spearman (commission)	0.676	0.728	0.740

When the BG/NBD and Gamma-Gamma outputs are combined into a single expected commission estimate, the resulting metric directly reflects the platform's revenue perspective, since marketplace income is structurally proportional to seller-generated GMV. MAE on commission increases from BRL 44.93 at 30 days to BRL 77.36 and BRL 111.32 at 60 and 90 days respectively, consistent with the accumulation of absolute error over longer windows. WAPE on commission ranges from 70.8% to 75.3%, indicating that precise seller-level commission forecasting remains challenging. This is an outcome primarily attributable to order volume uncertainty from the BG/NBD component, as the Gamma-Gamma model demonstrates superior stability on its own.

The Spearman rank correlation on commission, however, presents a more favorable picture: it improves from 0.676 at 30 days to 0.728 at 60 days and 0.740 at 90 days. Surpassing 0.7 at the 60-day horizon is a meaningful threshold, as it indicates that the combined model produces a reliable seller ranking by expected revenue contribution. From a business operations standpoint, decisions such as seller prioritization, support resource allocation, and engagement campaign targeting do not require exact commission forecasts; they require a trustworthy ordering of sellers by value potential. The Spearman results demonstrate that the combined model meets this requirement with increasing reliability as the horizon extends.

4) Comparative Analysis of Prediction Horizons:

5) K-means:

VI. CONCLUSION

REFERENCES

[1] P. S. Fader, B. G. S. Hardie, and K. L. Lee, "Counting Your Customers the Easy Way: An Alternative to the Pareto/NBD Model," *Marketing Science*, vol. 24, no. 2, pp. 275–284, 2005.

[2] D. C. Schmittlein, D. G. Morrison, and R. Colombo, "Counting Your Customers: Who Are They and What Will They Do Next?" *Management Science*, vol. 33, no. 1, pp. 1–24, 1987.

[3] P. S. Fader and B. G. S. Hardie, "The Gamma-Gamma Model of Monetary Value," technical note, Feb. 2013.

[4] P. S. Fader, B. G. S. Hardie, and K. L. Lee, "RFM and CLV: Using Iso-Value Curves for Customer Base Analysis," *Journal of Marketing Research*, vol. 42, no. 4, pp. 415–430, 2005.

[5] A. Mammadzada, E. Alasgarov, and A. Mammadov, "Application of BG/NBD and Gamma-Gamma Models to Predict Customer Lifetime Value for Financial Institution," in *Proc. IEEE 15th Int. Conf. on Application of Information and Communication Technologies (AICT)*, pp. 1–6, 2021.

[6] I.-K. Yeo and R. A. Johnson, "A New Family of Power Transformations to Improve Normality or Symmetry," *Biometrika*, vol. 87, no. 4, pp. 954–959, 2000.